

A Realized Variance for the Whole Day Based on Intermittent High-Frequency Data*

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Abstract

We consider the problem of deriving an empirical measure of daily *integrated variance* (IV) in the situation where high-frequency price data are unavailable for part of the day. We study three estimators in this context and characterize the assumptions that justify their use. We show that the optimal combination of the *realized variance* and squared overnight return can be determined, despite the latent nature of IV , and we discuss this result in relation to the problem of combining forecasts. Finally, we apply our theoretical results and construct four years of daily volatility estimates for the 30 stocks of the Dow Jones Industrial Average.

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1 Introduction

High-frequency data are increasingly being used in econometrics, because of the information they contain about population parameters. High-frequency data have successfully been used to address a wide range of issues, including macro economic aspects and the analysis of trading mechanisms.¹ But high-frequency data have primarily been used for estimation of financial volatility and the realized variance (RV) is a well known quantity that is constructed from high-frequency intraday returns. The RV is interesting because it can be used as a proxy for theoretical quantities such as the integrated variance (IV), see Andersen & Bollerslev (1998), Barndorff-Nielsen & Shephard (2002a, 2002b), and Meddahi (2002). However, in order to calculate a RV that spans a full day one needs high-frequency data for the entire 24 hours. While high-frequency data are available continuously for some exchange rates (brushing aside weekends) most equities are only actively traded for a fraction of the day, such as six to seven hours a day.

In this situation with *intermittent high-frequency data*, a substantial part of daily volatility may unfold during the inactive period. So we note that the squared return over the inactive period, $r_{1,t}^2$, may contain valuable information about daily volatility. In this paper, we discuss three estimators of daily volatility that are based on the squared close-to-open return, $r_{1,t}^2$, and a bias-adjusted realized variance for the open-to-close period, $RV_{2,t}$. We characterize assumptions that justify their use and explore ways to combine these two statistics, $r_{1,t}^2$ and $RV_{2,t}$ in an optimal manner. Using the mean squared error (MSE) criterion, we determine the optimal linear combination of $r_{1,t}^2$ and $RV_{2,t}$. This problem is made difficult by the fact that the true daily volatility is a latent variable. Fortunately, the problem is simplified considerably by restricting our attention to conditionally unbiased measures. For this class estimators we show that the MSE-optimal estimator has the smallest unconditional variance. The finding is quite useful because it makes it possible to determine the optimal weights empirically. This result relates a conditional MSE to an unconditional variance, so the result is different from the simple identity, $MSE = \text{variance} + \text{bias}^2$. In fact, the result has some interesting implications for forecast combination that we discuss in some details. We apply our results to the 30 stocks of the Dow Jones Industrial Average (DJIA), and construct 4 years of daily volatility measures for these equities.

Our estimator for the active period, $RV_{2,t}$, is based on the work of Hansen & Lunde (2003, 2006) who have shown that a simple Newey-West type correction can produce an unbiased measure of the volatility for the ‘open’ period. Our empirical results reveal several similarities

¹For example, Hamilton (1996) used *daily* Fed fund rates data to analyze the monetary transmission mechanism; Andersen, Bollerslev, Diebold & Vega (2003) use real-time quotations of exchange rates to analyze the effects of macroeconomic announcements; and Faust, Rogers, Swanson & Wright (2003) use high-frequency data to measure the impact of FOMC decisions. For analyses of trading mechanisms, see e.g. Engle & Lange (2001), Engle & Russell (2002), and Engle & Lunde (2003).

across equities, despite a great deal of variation in their level of volatility. About 80% of daily volatility occurs during the active period and this percentage is fairly constant across equities. Similarly, we find the correlation between the squared ‘overnight’ return, $r_{1,t}^2$, and the realized variance for the active period, $RV_{2,t}$, to be fairly constant across equities – typically in the neighborhood of 0.30. When we estimate the optimal combination of $r_{1,t}^2$ and $RV_{2,t}$ the latter is given a disproportionately larger weight, which is explained by the fact that $r_{1,t}^2$ is a relatively more noisy estimator than is $RV_{2,t}$.

A precise measure of daily volatility is useful for a range of applications. One is the evaluation of volatility models. For example, Andersen & Bollerslev (1998) have shown that a precise estimator of volatility, such as the RV , is needed in order to correctly assess the performance of ARCH-type models. Another important argument for using a precise measure of volatility is made by Hansen & Lunde (2005a). They show that a noisy measure of volatility can induce an inconsistent ranking of volatility models. This happens because the measurement errors can have a different impact on different models, unless the criterion function that is used for the evaluation satisfies certain conditions. See also Patton (2005) who characterize a class of loss function that are robust to this inconsistency problem. Building prediction models for volatility is another area where high-frequency based estimators have been shown to be valuable. For example, Andersen, Bollerslev & Meddahi (2004) have shown that a simple prediction model that is based on the RV , is almost as accurate as the true data generating model, see also Meddahi (2002) and Koopman, Jungbacker & Hol (2005). Furthermore, the integrated variance over multiple days is simply the sum of the daily integrated variances. Thus, daily estimates of volatility can be added up to form an estimator of multi-day IV .

This paper is organized as follows. Section 2 contains a brief discussion of the realized variance, market microstructure noise, and two bias corrected RV s. In Section 3 we study the problem of estimating daily volatility from intermittent high-frequency data. Specifically, we consider ways to combine the bias corrected RV for the active part of the day with the squared ‘over-night’ return, and we determine the optimal unbiased linear combination of the two. Section 4 presents our empirical analysis, where our theoretical results are applied to the DJIA stocks. We construct four years of daily volatility data that are available from the authors upon request. The assumptions that underlie our estimators are put to test in an extensive analysis of their validity. Section 5 contains a summary and some concluding remarks, and an Appendix presents the proofs of our theoretical results.

2 Realized Variance: Definitions and Notation

Let $\{p^*(t)\}_{t \in [0, \infty)}$ denote the logarithmic efficient price process and assume that dividends (if any) are accumulated in $p^*(t)$. The observed price process, p , may differ from the efficient price process, p^* , due to market microstructure noise, and we accommodate this situation by

introducing a noise process, u , that is defined from the identity, $p = p^* + u$. We let integer values of t correspond to market closing-times and use a trading day as the unit of time, such that $r_t \equiv p(t) - p(t-1)$ is the (close-to-close) return on day t .

We shall assume that the price process satisfies the stochastic differential equation, $dp^*(t) = \mu(t)dt + \sigma(t)dw(t)$, where w is a standard Brownian motion and μ and σ are ‘smooth’ time-varying (random) functions that are independent of w . This allows us to define the integrated variance,

$$IV_{[a,b]} \equiv \int_a^b \sigma^2(t)dt,$$

that is our object of interest.

The RV is an empirical estimate of the IV that is constructed from intraday returns. Given the times, $a = t_0 < t_1 < \dots < t_m = b$, at which the price is observed, we define the *realized variance at sampling* $\Xi = \{t_0, \dots, t_m\}$ by

$$RV_{[a,b]}^\Xi \equiv \sum_{i=1}^m \{p(t_i) - p(t_{i-1})\}^2. \quad (1)$$

We call Ξ a partition of $[a, b]$ and note that $RV_{[a,b]}^\Xi$ is specific to the partition. For the special case where intraday returns are equidistant in calendar time, such that $t_i = t_{i-1} + \Delta$ for $i = 1, \dots, m$, where $\Delta \equiv (b - a)/m$ and $t_0 = a$, we define the intraday returns, $y_{i,m} = p(t_i) - p(t_{i-1})$, and the *regular realized variance* over the interval $[a, b]$,

$$RV_{[a,b]}^{(m)} \equiv \sum_{i=1}^m y_{i,m}^2. \quad (2)$$

Equidistant price observations must typically be interpolated from transaction prices or quotations, such as the *previous-tick* and *linear interpolation* methods that were introduced by Wasserfallen & Zimmermann (1985) and Andersen & Bollerslev (1997), respectively. See Dacorogna, Gencay, Müller, Olsen & Pictet (2001, sec. 3.2.1) for a discussion of the two methods, and see Hansen & Lunde (2003, 2006) for a theoretical argument that favors the previous-tick method. Our empirical results are based on the previous-tick method.

2.1 Bias and Bias Correcting the RV

It is well known that the standard measure of RV , (1), suffers from a bias problem that is due to autocorrelation in the intraday returns. The autocorrelation is caused by market microstructure effects such as: bid-ask bounces, non-synchronous trading, rounding errors, and misrecordings, see e.g., Andreou & Ghysels (2002), Bai, Russell & Tiao (2004), and Oomen (2002).

The autocorrelation in intraday returns becomes more of an issue as the sample-frequency, m , increases. So it is common to use a moderately large value for m , and an m that corresponds to five-minute intraday returns is often used in practice, see e.g., Andersen & Bollerslev (1997).

The optimal sampling frequency under independent market microstructure noise² was derived by Bandi & Russell (2004) and Zhang, Mykland & Aït-Sahalia (2005), and the optimal sampling frequency is often found to be between one and five minutes, see Bandi & Russell (2004), Oomen (2004), and Hansen & Lunde (2006).

An obvious drawback of sampling at low frequencies (to avoid the bias) is that this approach discards information, such that the resulting estimator may be inefficient. In fact, Hansen & Lunde (2006) show that a correction for the first-order autocorrelation can eliminate the bias and reduce the RMSE by as much as 50%. This results is derived under the ‘independent’ noise assumption, see footnote 2. Under more realistic assumptions where the noise may be time-dependent and correlated with efficient returns, Hansen & Lunde (2006) show that kernel-based estimators that utilize higher-order autocovariances can eliminate the bias of the RV . Specifically they show that

$$RV_{AC[a,b]}^{(m)} \equiv \sum_{i=1}^m y_{i,m}^2 + 2 \sum_{h=1}^q \frac{m}{m-h} \sum_{i=1}^{m-h} y_{i,m} y_{i+h,m},$$

is conditionally unbiased for $IV_{[a,b]}$ under mild assumptions, see Hansen & Lunde (2006) for details. The estimator $RV_{AC[a,b]}^{(m)}$ achieves unbiasedness by an upwards scaling of the empirical autocovariances. The h th autocovariance is scaled by $m/(m-h)$ to compensate for the “missing” autocovariance terms, such as $y_{m,m}y_{m+h,m}$ that requires intraday returns from outside the $[a, b]$ interval. The upwards scaling has the drawback that it increases the variance of the estimator. For this reason we shall employ the Bartlett kernel and define the estimator,

$$RV_{NW[a,b]}^{(m)} \equiv \sum_{i=1}^m y_{i,m}^2 + 2 \sum_{h=1}^q \left(1 - \frac{h}{q+1}\right) \sum_{i=1}^{m-h} y_{i,m} y_{i+h,m},$$

that we refer to as the Newey-West modified realized variance. This estimator has the advantage that it is guaranteed to be non-negative, see Newey & West (1987), a property that is not shared by $RV_{AC[a,b]}^{(m)}$. Furthermore, Barndorff-Nielsen, Hansen, Lunde & Shephard (2004) have recently shown that $RV_{NW}^{(m)}$ is almost identical to the subsample-based estimator of Zhang et al. (2005), which is consistent for IV when the noise is of the ‘independent’ type, see Zhang et al. (2005) for details. For other approaches to bias correcting the RV , see Andersen, Bollerslev, Diebold & Ebens (2001), Bollen & Inder (2002), Bandi & Russell (2004), and Oomen (2004).

3 An Optimal RV for the Whole Day

To simplify our notation, we write $IV_{1,t} \equiv IV_{[t-1,t-\Delta_o]}$ and $IV_{2,t} \equiv IV_{[t-\Delta_o,t]}$ to represent the IV s over the inactive and active part of the day, respectively. Similarly, we define the close-to-open return, $r_{1,t} \equiv p(t - \Delta_o) - p(t - 1)$, and the open-to-close return $r_{2,t} \equiv p(t) - p(t - \Delta_o)$,

²Independent refers to absence of time-dependence and independence between noise and the efficient price.

such that $r_t = r_{1,t} + r_{2,t}$. For the open-to-close period where we have high-frequency intraday returns, we let $RV_{2,t} = RV_{[t-\Delta_o, t]}$ represent some RV -measure for the active part of day. In our empirical analysis we use $RV_{NW[t-\Delta_o, t]}^{(m)}$ as our measure for $RV_{2,t}$. Our notation for the whole day is $IV_t \equiv IV_{[t-1, t]}$.

Next, we consider three ways to extend $RV_{2,t}$ to a measure of volatility for the full day. The first approach is a simple scaling of the $RV_{2,t}$ and the second approach is to add the squared overnight return, $r_{1,t}^2$, to $RV_{2,t}$. Finally, we consider linear combinations of $r_{1,t}^2$ and $RV_{2,t}$ and construct our third estimator of IV_t by choosing the weights that minimizes the mean squared error. Our setup is illustrated in Figure 1.

[Figure 1 about here]

3.1 Scaling Estimator of IV

The idea behind this estimator is to scale $RV_{2,t}$ by a constant, such that the resulting estimator has the correct expected value. Thus, we consider

$$RV_t^{\text{scale}} \equiv \hat{\delta} \cdot RV_{2,t}, \quad (3)$$

where $\hat{\delta} = \sum_{t=1}^n (r_t - \bar{r})^2 / \sum_{t=1}^n RV_{2,t}$ is a consistent estimator of $\delta \equiv E[IV_t] / E[RV_{2,t}]$, and where $\bar{r} = n^{-1} \sum_{t=1}^n r_t$.³ This estimator has previously been applied by Hansen & Lunde (2005b), and a similar estimator, $\tilde{\delta} \equiv (\sum_{t=1}^n r_{2,t}^2 + \sum_{t=1}^n r_{1,t}^2) / \sum_{t=1}^n r_{2,t}^2$, was concurrently suggested by Martens (2002) and Koopman et al. (2005). The latter is expected to be less precise than $\hat{\delta}$, because $\text{var}(r_{2,t}^2)$ is several times larger than is $\text{var}(RV_{2,t})$. Fleming, Kirby & Ostdiek (2003) take a dynamic scaling approach, $\check{\delta}_t \equiv (\sum_{s=1}^t \rho^s r_{t-s}^2) / \sum_{s=1}^t \rho^s RV_{2,t-s}$, which is similar to $\hat{\delta}$, in the sense that it is also based on a RV for the open period. Here $\rho \in (0, 1)$ controls the exponential weighting of lagged values of r_t^2 and $RV_{2,t}$.

While scaling the $RV_{2,t}$ appears to be a sensible thing to do, it is interesting to characterize the conditions that justify a simple scaling. We address this issue in Theorem 1 that formulates a set of conditions that justifies RV_t^{scale} as a proper estimator of daily volatility. The theorem is based on two assumptions. Our first assumption ensures that a (particular) scaling of $RV_{2,t}$ is informative about IV_t , and our second assumption ensures that the scaling coefficient can be estimated consistently by incorporating information from an increasing number of days. To simplify notation, we use $E_\sigma(\cdot) \equiv E(\cdot | IV_t)$ to denote the conditional expectation.

Assumption 1 *We assume that:*

$$(i) \quad E_\sigma(IV_{2,t}) = \delta_o IV_t;$$

³We expect $\delta < 1/\Delta_o$ as relatively more information arrives during the hours that the market is open, see French & Roll (1986) and Baillie & Bollerslev (1989).

- (ii) $E_\sigma(\delta_{b_2}RV_{2,t} - IV_{2,t}) = 0$; and
- (iii) $E_\sigma(\delta_{b_1}r_{1,t}^2 - IV_{1,t}) = 0$.

The first part of Assumption 1 states that a fixed proportion of the daily integrated variance occurs during the active period. This does not require that the underlying stochastic process for the active period is identical to that of the inactive period. For example, the variance of an unobserved 1-minute return during the inactive period may be very different from the variance of a 1-minute return during the active period. This assumption also allows the ‘weekend’ volatility (Friday close to Monday open) to be larger, say, than that for other close-to-open periods, provided that the open-to-close volatility over the weekend deviates from other weekdays in a proportional way. Nevertheless, (i) is a restrictive assumption and our simple scaling estimator relies heavily on its validity, so we shall scrutinize the validity of this assumption in our empirical analysis. (ii) requires the conditional ‘bias’ of $RV_{2,t}$ be proportional to $IV_{2,t}$. This requirement is clearly satisfied for all RV -measures that are conditionally unbiased, and the assumption can also be shown to hold for certain biased estimators under suitable assumptions. For example, when market microstructure is of the independent type with constant variance then Fang (1996) has shown that the bias of $RV^{(m)}$ is proportional to the number of intraday returns, m , see also Bandi & Russell (2004) and Zhang et al. (2005). Consider the RV that is based on intraday returns, which are sampled every k th transaction and denote this estimator by $RV_{2,t}^{(kticks)}$. If the (expected) number of transactions per day is proportional to the level of volatility, $IV_{2,t}$, then the bias of $RV_{2,t}^{(kticks)}$ is proportional to $IV_{2,t}$, because the number of intraday returns that are used to compute $RV_{2,t}^{(kticks)}$ is (on average) proportional to both $IV_{2,t}$ and the bias of $RV_{2,t}^{(kticks)}$. Another framework that would make the bias of $RV_{2,t}^{(m)}$ proportional to $IV_{2,t}$ is given in the following example.

Example 1 Let $p_i = p_i^* + u_i$, $i = 0, \dots, m$, where p_i is the observed price, p_i^* is the latent efficient price, and u_i is a measurement error (market microstructure noise). Thus, $E(y_i^2) = E(y_i^* + u_i - u_{i-1})^2$, where $y_i^* \equiv p_i^* - p_{i-1}^*$, $i = 1, \dots, m$ are the efficient intraday returns that are assumed to be uncorrelated. Suppose that the noise is given by $u_i = \alpha y_i^*$, such that the full impact of a change in p_i^* is not transparent in p_i immediately (whenever $-1 < \alpha < 0$). Then

$$E[y_i^2] = E[(1 + \alpha)y_i^* - \alpha y_{i-1}^*]^2 = (1 + \alpha)^2 E(y_i^{*2}) + \alpha^2 E(y_{i-1}^{*2}),$$

and

$$E\left[\sum_{i=1}^m y_i^2\right] \approx IV + \underbrace{2\alpha(1 + \alpha)IV}_{\text{bias}},$$

such that the bias of $RV_{[a,b]}^{(m)}$ is proportional to the IV in this case.

It is also easy to verify that

$$E\left[\sum_{i=1}^{m-1} y_i y_{i+1}\right] \approx -\frac{m-1}{m}\alpha(1+\alpha)IV,$$

which shows that $RV_{NW[a,b]}^{(m)}$ is slightly biased, with a bias that is proportional to IV .

Finally, Assumption 1 (iii) requires that the squared over-night return, $r_{1,t}^2$, is conditionally proportional to the ‘overnight-IV’ (but not necessarily unbiased). The last part of the assumption, (iii), is not used in our analysis of the scaling estimator. This requirement is needed in our subsequent analysis where the squared over-night return is incorporated in our estimator.

Assumption 2 Assume that $\{RV_{2,t}\}$, $\{r_t\}$, and $\{r_t^2\}$ satisfy a law of large numbers as $n \rightarrow \infty$.

With Assumptions 1 and 2 in place we are ready to formulate the result that justifies the scaling estimator.

Theorem 1 Given Assumption 1.i-ii it follows that $E_\sigma(\delta RV_{2,t}) = IV_t$, where $\delta \equiv \delta_{b_2}/\delta_o$. Suppose further that Assumption 2 holds and that $\mu(t) = \mu$ is constant. Then $\hat{\delta} \xrightarrow{P} \delta$, as $n \rightarrow \infty$, such that $RV_t^{scale} - \delta RV_{2,t} = o_p(1)$ as $n \rightarrow \infty$.

Note that $RV_{2,t}$ is not required to be unbiased for $IV_{2,t}$, as long as the conditional bias is proportional to $IV_{2,t}$. This is convenient in the present context, because the $RV_{NW}^{(m)}$ may be slightly biased for the IV when market microstructure noise is time-dependent and correlated with the efficient returns. We adopt this estimator, $RV_{NW}^{(m)}$, in our empirical analysis.

3.2 Incorporating the ‘Over-Night’ Return

In addition to the high-frequency intraday returns (during open-to-close) we also have the ‘overnight’ return, $r_{1,t}$, available for our estimation. Thus, instead of scaling the $RV_{2,t}$ we can incorporate $r_{1,t}$ in our measure of daily volatility. The simplest way to include $r_{1,t}$ is through the *irregular RV* for the full day, see (1). This estimator is given by

$$RV_t^{+on} \equiv r_{1,t}^2 + RV_{2,t}, \quad (4)$$

where the two terms that define RV_t^{+on} can be thought of as estimators of the IV during the active period and the inactive period, respectively. An obvious drawback of this measure, RV_t^{+on} , is that the squared over-night return, $r_{1,t}^2$, is likely to be a noisy estimator of $IV_{1,t}$. Just as the squared *interday* returns, r_t^2 , is a noisy estimator of IV_t . So this estimator defies the purpose of using high-frequency data, which is to obtain a precise estimate of the volatility.

The two estimators, RV_t^{scale} and RV_t^{+on} , are linear combinations of $r_{1,t}^2$ and $RV_{2,t}$, using the weights $(0, \delta)$ and $(1, 1)$, respectively. More generally we could consider the class of estimators, given by all linear combinations of $r_{1,t}^2$ and $RV_{2,t}$. So we define

$$RV_t(\omega) \equiv \omega_1 r_{1,t}^2 + \omega_2 RV_{2,t}, \quad \text{where } \omega \equiv (\omega_1, \omega_2)',$$

and consider the problem

$$\min_{\omega \in \Omega} E [RV_t(\omega) - IV_t]^2, \quad (5)$$

for some suitable class of estimators, $\Omega \subset \mathbb{R}^2$. Unfortunately, it is not possible to obtain an empirical quantity, such as

$$n^{-1} \sum_{i=1}^n [RV_t(\omega) - IV_t]^2,$$

that can approximate the objective function in (5), because IV_t is a latent variable. However, by restricting attention to conditionally unbiased estimators we achieve a major simplification that enables us to reformulate the problem in a way that does not require the evaluation of (5). This is achieved by showing that a pseudo objective that can be solved empirically, has the same solution as (5). This result is formulated in Corollary 3 that follows from the following lemma.

Lemma 2 *Let Y be a real random variable and let $\{X_\gamma : \gamma \in \Gamma\}$ be a class of real random variables. If $E(X_\gamma|Y) = Y$ for all $\gamma \in \Gamma$ then*

$$\arg \min_{\gamma \in \Gamma} E(X_\gamma - Y)^2 = \arg \min_{\gamma \in \Gamma} \text{var}(X_\gamma).$$

We omit the proof that amounts to verifying the identity $E(X_\gamma - Y)^2 = \text{var}(X_\gamma) - \text{var}(Y)$, whenever $Y = E(X_\gamma|Y)$. Lemma 2 is more substantive than it may appear to be at first sight. In the special case where Y is a constant $Y = c \in \mathbb{R}$, the reader will be familiar with the well known result, $E(X_\gamma - c)^2 = \text{var}(X_\gamma) + \text{bias}^2$, where $\text{bias} = E(X_\gamma) - c$. So the result of the lemma is quite obvious in this special case. The essence of Lemma 2 is buried in the fact that Y need not be constant (and need not be observable). While $E(X_\gamma - Y)^2$ cannot be estimated when Y is a latent random variable, we will often be able to estimate $\text{var}(X_\gamma)$. So Lemma 2 yields a useful tool for comparing (and combining) conditionally unbiased proxies for Y . Setting $IV_t = Y$, $\omega = \gamma$, and $RV_t(\omega) = X_\omega$ in Lemma 2 yields the following corollary.

Corollary 3 *Let Ω be such that $E_\sigma[RV_t(\omega) - IV_t] = 0$ for all $\omega \in \Omega$. The pseudo objective*

$$\min_{\omega \in \Omega} \text{var}[RV_t(\omega)], \quad (6)$$

has the same solution as (5).

In the following we restrict our attention to estimators that are conditionally unbiased for IV_t , such that we can invoke the result of Corollary 3. While this clearly excludes a large class of (conditionally biased) estimators, there does not appear to be a straight forward way to solve (5) empirically for a richer class of estimators without making additional assumptions.

The conditionally unbiased linear combinations are relatively easy to determine due to the following Lemma, where we have defined $\mu_0 \equiv E(IV_t)$, $\mu_1 \equiv E(r_{1,t}^2)$, and $\mu_2 \equiv E(RV_{2,t})$.

Lemma 4 *Given Assumption 1, we have $E_\sigma[RV_t(\omega)] = IV_t$ for all ω that satisfy $\omega_1\mu_1 + \omega_2\mu_2 = \mu_0$.*

By combining Corollary 3 and Lemma 4 we now have that the optimal (unbiased) linear estimator, $RV_t(\omega^*)$, is given as the solution to the constrained optimization problem:

$$\min_{\omega_1, \omega_2} \text{var}(\omega_1 r_{1,t}^2 + \omega_2 RV_{2,t}), \quad \text{s.t. } \omega_1\mu_1 + \omega_2\mu_2 = \mu_0. \quad (7)$$

The solution is given by the following theorem.

Theorem 5 *Let $\eta_1^2 \equiv \text{var}(r_{1,t}^2)$, $\eta_2^2 \equiv \text{var}(RV_{2,t})$, and $\eta_{12} \equiv \text{cov}(r_{1,t}^2, RV_{2,t})$. The solution to (5) is given by*

$$\omega_1^* \equiv (1 - \varphi) \frac{\mu_0}{\mu_1} \quad \text{and} \quad \omega_2^* = \varphi \frac{\mu_0}{\mu_2}, \quad (8)$$

where φ is a relative importance factor, defined by

$$\varphi \equiv \frac{\mu_2^2 \eta_1^2 - \mu_1 \mu_2 \eta_{12}}{\mu_2^2 \eta_1^2 + \mu_1^2 \eta_2^2 - 2\mu_1 \mu_2 \eta_{12}}. \quad (9)$$

This solution is intuitive and is particularly simple to interpret if $\eta_{12} = 0$. In this special case we have that $\omega_1^*/\omega_2^* = \frac{\mu_2/\mu_1}{\eta_2^2/\eta_1^2}$, which shows that an increase in volatility during the active period (relative to the inactive period) has a positive impact on the relative weight $\partial(\omega_1^*/\omega_2^*)/\partial(\mu_2/\mu_1) > 0$, whereas the opposite is the case for an increase in the relative noise, $\partial(\omega_1^*/\omega_2^*)/\partial(\eta_2^2/\eta_1^2) < 0$.

The result of Theorem 5 is straight forward to implement in practice, because the quantities, μ_1 , μ_2 , η_1 , η_2 , and η_{12} , can be estimated by simple sample averages that are consistent under suitable regularity conditions that are similar to those we stated in Assumption 2.

3.3 An Interesting Connection to Forecast Combination

The problem of combining $r_{1,t}^2$ and $RV_{2,t}$ is akin to that of combining forecasts, as was pointed out by an anonymous referee. Bates & Granger (1969) analyzed the problem of combining unbiased forecasts and in the present context we may interpret

$$\frac{\mu_0}{\mu_1} r_{1,t}^2 \quad \text{and} \quad \frac{\mu_0}{\mu_2} RV_{2,t},$$

as two unbiased ‘forecasts’ of IV_t . The optimal weighting implied by (9) is therefore closely related to a result by Bates & Granger (1969), see also Diebold & Lopez (1996).

This connection (and an important difference) is interesting to spell out in details. Let $\hat{y}_1$ and $\hat{y}_2$ be two unbiased forecasts of y , such that the forecast errors, $e_1 \equiv y - y_1$ and $e_2 \equiv y - y_2$ satisfies $E(e_i) = 0$, $i = 1, 2$. The optimal weights in Bates & Granger (1969) are given by

$$\frac{\sigma_{e_2}^2 - \sigma_{e_1 e_2}}{\sigma_{e_1}^2 + \sigma_{e_2}^2 - 2\sigma_{e_1 e_2}} \quad \text{and} \quad \frac{\sigma_{e_1}^2 - \sigma_{e_1 e_2}}{\sigma_{e_1}^2 + \sigma_{e_2}^2 - 2\sigma_{e_1 e_2}}, \quad (10)$$

where

$$\sigma_{e_1}^2 \equiv \text{var}(e_1), \quad \sigma_{e_2}^2 \equiv \text{var}(e_2), \quad \text{and} \quad \sigma_{e_1 e_2} \equiv \text{cov}(e_1, e_2).$$

In our framework y is a latent variable ($= IV_t$), so the empirical versions of $\sigma_{e_1}^2$, $\sigma_{e_2}^2$, and $\sigma_{e_1 e_2}$, cannot be computed. Instead we use estimates of the quantities,

$$\sigma_{\hat{y}_1}^2 \equiv \text{var}(\hat{y}_1), \quad \sigma_{\hat{y}_2}^2 \equiv \text{var}(\hat{y}_2), \quad \text{and} \quad \sigma_{\hat{y}_1 \hat{y}_2} \equiv \text{cov}(\hat{y}_1, \hat{y}_2),$$

and define the optimal weights to be

$$\frac{\sigma_{\hat{y}_2}^2 - \sigma_{\hat{y}_1 \hat{y}_2}}{\sigma_{\hat{y}_1}^2 + \sigma_{\hat{y}_2}^2 - 2\sigma_{\hat{y}_1 \hat{y}_2}} \quad \text{and} \quad \frac{\sigma_{\hat{y}_1}^2 - \sigma_{\hat{y}_1 \hat{y}_2}}{\sigma_{\hat{y}_1}^2 + \sigma_{\hat{y}_2}^2 - 2\sigma_{\hat{y}_1 \hat{y}_2}}. \quad (11)$$

How can we reconcile these two definitions of the optimal weights? The weights in (10) and (11) must be identical, because they are (unique) solutions to the same problem. The equivalence is a corollary of Lemma 2, but the result can also be shown directly by noting that the assumed unbiasedness of $\hat{y}_1$ and $\hat{y}_2$ (conditional on y) implies that

$$\text{cov}(e_i, e_j) = \text{cov}(\hat{y}_i, \hat{y}_j) - \text{var}(y), \quad i, j = 1, 2.$$

This result is interesting because it suggests that the result of Bates & Granger (1969) can be generalized to the situation where the object of interest is a latent variable. However, there is one important difference. In our framework it is assumed that $E(e_i|y) = 0$, $i = 1, 2$. The analogous conditional moment conditions in Bates & Granger (1969) are different, as they are given by, $E(e_i|\mathcal{F}) = 0$, where $\mathcal{F}$ is some σ -algebra to which $\hat{y}_1$ and $\hat{y}_2$ are adapted.

4 Empirical Application to DJIA Equities

We will apply the result of the previous section to the 30 equities of the DJIA. These are listed in Table 1 along with their acronyms and some descriptive statistics.

[Table 1 about here]

Our sample period contains $n = 986$ trading days that span the period from January 29, 2001 to December 31, 2004. This period was chosen because it is the longest sample period that is currently available after the price-tick was reduced to *one cent* from *a sixteenth of a dollar*. A constant price-tick in the sample period is desirable because the price-tick is likely to influence the magnitude of market microstructure noise, hence the bias of $RV_{2,t}$. The data are quotations from the NYSE and NASDAQ that were extracted from the Trade and Quote (TAQ) database. The raw data were filtered for outliers and we discarded transactions outside the period from 9:30am to 4:00pm. Intraday returns were constructed for both bid and ask prices using the previous-tick interpolation method, see Wasserfallen & Zimmermann (1985).

The overnight return was defined as the log-difference between the first price (mid quote) of the day and the last price (mid quote) of the preceding day. The overnight returns were corrected for stock splits and dividend payments.

[Figure 2 about here]

We calculated $RV_{NW}^{(m)}$ for both bid and ask prices, and the average of these two Newey-West modified RVs were used as our definition of $RV_{2,t}$. The number of autocovariances, $q = q_m$, were such that q_m intraday returns span a 10 minute period. This window effectively reduces the bias problem that is very pronounced for the standard RV. This is evident from Figure 2 where we have plotted volatility signature plots for both $RV^{(m)}$ and the Newey-West modified RV, using return data from AA, MSFT, GE, and PG. A volatility signature plot is a powerful way to visualize potential bias problems, and such plots first appeared in Fang (1996) and in Andersen, Bollerslev, Diebold & Labys (2000) who named these plots, *volatility signature plots*. Figure 2 reveals a pronounce bias of the $RV^{(m)}$ in each of the sub-plots. The bias is caused by market microstructure noise and it tends to increase with the sampling frequency. Hansen & Lunde (2006) have established the link between general forms of market microstructure noise and the bias. It is well understood that the variance of the market microstructure noise causes a positive bias, see Fang (1996), Bandi & Russell (2004), and Zhang et al. (2005). Interestingly, Hansen & Lunde (2006) have shown that a correlation between noise and efficient returns leads to a second bias component that has the same sign as the correlation. So this bias component can go in either direction. In fact, if the correlation is sufficiently negative the second bias component can more than offset the first component. Thus the total bias may be negative, as is often seen when $RV^{(m)}$ is based on mid-quotes. In the present situation, where we use bid and ask prices the bias is positive for AA, GE, and PG, whereas the total bias is negative for the highly liquid stock, MSFT. The important observation to make in relation to the present problem, is that the signature plots for $RV_{NW}^{(m)}$ are quite flat. So this estimator does not appear to suffer from a severe bias problem. In our empirical analysis we will estimate $RV_{NW}^{(m)}$ using an m that is such that each of the intraday returns span 10 seconds long period of time. Thus in the notation of Hansen & Lunde (2006) or high-frequency based estimator for the open-to-close period is defined by

$$RV_{2,t} \equiv \frac{1}{2}RV_{NW_{60},ASK}^{(10 \text{ sec})} + \frac{1}{2}RV_{NW_{60},BID}^{(10 \text{ sec})}.$$

The importance factor, φ , and the weights, ω_1^* and ω_2^* , were estimated by plugging estimates of μ_0 , μ_1 , μ_2 , η_1^2 , η_2^2 , and η_{12} into the expressions (8) and (9). Our unrestricted estimates were found to be strongly influenced by the most extreme values for $r_{1,t}^2$. So initially we employed a bivariate log-Gaussian model for $(r_{1,t}^2, RV_{2,t})$, hoping that this model would be less sensitive to “outliers”. Unfortunately, the log-Gaussian distribution turned out to be a poor approximation of the distribution of $r_{1,t}^2$, and the quasi maximum likelihood estimates from this model were

found to be unreliable. In the end, we settled on a simpler procedure where about 1% of the observations were classified as outliers and omitted from the estimation (but not the subsequent evaluation).⁴ While overnight returns tend to be relatively small, some are very large because the price “jumps” in the close-to-open period – typically as a response to important news that arrives “overnight”. To reduce the influence of such “outliers” in a manageable way for all 30 stocks, we simply removed the seven largest observation of $r_{1,t}^2$ and the three largest $RV_{2,t}$ in our estimation of the optimal weights. The parameters were then estimated using standard formulae, see Table 2 that also contains the estimated values of μ_0 , μ_1 , μ_2 , η_1^2 , η_2^2 , η_{12} , and φ .

[Table 2 about here]

There are several interesting observations to be made from Table 2. First, the column with the estimates of μ_0 reveals a substantial variation in the level of volatility across the equities. However, the ratio of volatility that occurs during the active (or inactive) period is quite stable across equities. The volatility ratio of ‘active’ to ‘inactive’, $\hat{\mu}_2/\hat{\mu}_1$, is about 4, which suggests that about 20% of daily volatility occurs during the inactive period. Interestingly, it is three technology stocks, HPQ, IBM, and INTC, that have the smallest ratios for, $\hat{\mu}_2/\hat{\mu}_1$, whereas more ‘industrial’ equities, IP, MMM, and PG, have the three largest ratios. Thus, the tech stocks appear to have a relatively larger proportion of their volatility occurring during the close-to-open period. Second, there is a great deal of dispersion across assets of the *volatility-of-volatility*, η_1^2 and η_2^2 , and the ratio of the two, $\hat{\eta}_2^2/\hat{\eta}_1^2$. However, the correlations are surprisingly similar, as these lie between .25 and .35 in most cases. Third, we find it very interesting that the estimates of φ are close to one in most cases. The minimum value is 0.858 (JNJ) and the maximum value is 0.986 (EK). Here it is worth to recall that RV_t^{scale} and RV_t^{on} correspond to $\varphi = 1$ and $\varphi = 1/2$, respectively (apart from a small bias correction of RV_t^{on}). So the scaling estimator is far closer to the optimal estimator than is the irregular RV in this metric. That φ is close to one reflects the fact that $RV_{2,t}$ is a far more precise estimator of $IV_{2,t}$ than is $r_{1,t}^2$ of $IV_{1,t}$. This aspect is also the main reason that the ‘optimal’ estimator

$$RV_t^* \equiv \hat{\omega}_1 r_{1,t}^2 + \hat{\omega}_2 RV_{2,t},$$

places disproportionately more weight on $RV_{2,t}$.

4.1 Further Refinements

While RV_t^* aspires to be the best linear combination of $r_{1,t}^2$ and $RV_{2,t}$, this estimator can be improved upon by incorporating additional information. In this subsection we discuss some avenues for improving our estimator, RV_t^* , and we construct a second set of daily volatility measures based on one of these methods.

⁴Outliers were defined to be the days that had one of the seven largest observation of $r_{1,t}^2$, or one of the three largest $RV_{2,t}$.

One improvement can be achieved by making stronger assumptions about the efficient prices, p^* , and the noise process, u . Such assumptions imply a particular structure that may be utilized in the estimation of $IV_{2,t}$. For example, by specifying a fully parametric diffusion model for p^* , see e.g. Aït-Sahalia, Mykland & Zhang (2005) and Jungbacker & Koopman (2004). Another avenue for improvements that is more specific to the present context, is to assume a structure across the panel of equities. This may be used to improve the estimators of ω_1^* and ω_2^* . For example, one might go as far and assume that ω_1^* and ω_2^* are identical for certain groups of equities. Making assumptions about the dynamics of IV_t (across days) is a third and very promising way to improve the daily estimates. Assumptions of this kind can be motivated by the high persistence of daily volatility that is observed empirically. This approach makes it possible to incorporate information from neighboring days when estimating IV_t . For example by applying filtering or smoothing methods to $RV_t^{(m)}$, as analyzed and discussed in Maheu & McCurdy (2002), Barndorff-Nielsen, Nielsen, Shephard & Ysusi (1996), and Koopman et al. (2005).

[Figure 3 about here]

The Kalman filter is an effective way to smooth our daily estimators. We illustrate this by employing a simple Kalman filter to $\log(RV_t^*)$. Applying the Kalman filter to $\log(RV_t^*)$ rather than the level, RV_t^* , has two advantages. First, the implicit distributional assumptions (Gaussian) are closer to being satisfied. Second, the filtered volatility, $\widetilde{RV}_t^* = \exp(\widehat{IV}_t)$, is guaranteed to be positive, where $\widehat{IV}_t$ represents the fitted state variables. We also apply a simple Kalman filter to $\log(RV_{2,t})$, such that we construct a total of four volatility series: RV_t^* and $\widetilde{RV}_t^*$ are the daily quantities whereas $RV_{2,t}$ and $\widetilde{RV}_{2,t}^*$ are measures of volatility for the active part of the day. These time series are plotted in Figure 3, where the two filtered series, $\widetilde{RV}_t^*$ and $\widetilde{RV}_{2,t}^*$ where obtained using EViews' implementation of the Kalman filter that induces an ARMA(1,1) specification. To conserve space we only present these time-series for a single stock, AA, over a two year period (2003 and 2004). The daily quantities, RV_t^* and $\widetilde{RV}_t^*$, are plotted in the upper panel and the active-period quantities, $RV_{2,t}$ and $\widetilde{RV}_{2,t}^*$, are plotted in the lower panel. The lower panel shows that $RV_{2,t}$ is smoothen quite a bit by the filter. Thus when the Kalman filter is applied to the daily quantities, RV_t^* , $t = 1, \dots, n$, it does more than a simply smoothing of the noisy squared overnight returns, $r_{1,t}^2$. The four time series for each of the 30 equities are available from the authors upon request.

4.2 Tests of Identifying Assumptions

The latent nature of volatility made it necessary for us to make some fairly strong assumption about the ratio of volatility that occurs during the active and inactive periods.⁵ The scaling

⁵The distinct features of volatility during the inactive period have been analyzed in several paper. For example, Lockwood & Linn (1990) showed that the volatility is greater during the active period than the inactive period, and

estimator, RV_t^{scaled} , and the optimally combined estimator, RV_t^* , critically rely on the validity of Assumption 1.i, that states that a fixed proportion of daily volatility occurs during the active period. Because this assumption is key for our empirical results, we shall scrutinize its validity in a number of ways. Smoothing induces a particular form of autocorrelation. Thus in order to avoid the complication that the smoothing causes, we shall apply our tests to the raw time series, $RV_{2,t}$ and RV_t^* .

[Figure 4 here]

Figure 4 is a visual inspection of Assumption 1. The figure contains scatter plots of $\log(r_{1,t}^2)$ against $\log(RV_{2,t})$ for some selected equities.⁶ Each plot contains three regression lines that are estimated from the first half, second half, and the full sample. We have used different symbols for the two subsamples, Δ for the first subsample and $\circ$ for the second subsample. Note that the level of volatility was generally higher in the second subsample. The three regression lines are statistically indistinguishable and this can be viewed as an informal test of Assumption 1, because the assumption implies $RV_{2,t} = c_1 IV_t(1 + \varepsilon_t)$ and $r_{1,t}^2 = c_2 IV_t(1 + \nu_t)$ for some ‘error’ terms, ε_t and ν_t , that satisfies $E(\varepsilon_t|IV_t) = E(\nu_t|IV_t) = 0$. Thus, the ratio

$$\frac{r_{1,t}^2}{RV_{2,t}} = \frac{c_2}{c_1} \frac{1 + \nu_t}{1 + \varepsilon_t},$$

does not involve IV_t , and by the Taylor expansion, $\log(1 + x) = x + O(x^2)$, we note that

$$\log(r_{1,t}^2/RV_{2,t}) \simeq c + \nu_t - \varepsilon_t, \quad (12)$$

such that $E[\log(r_{1,t}^2/RV_{2,t})|IV_t] \simeq c + 0$, for some constant $c \in \mathbb{R}$.

Although Figure 4 does not yield noticeable evidence against Assumption 1, these scatter plots do not address all aspects of the assumption. Importantly, the conditional moment conditions, $E(\nu_t|IV_t) = 0$ and $E(\varepsilon_t|IV_t) = 0$, are not addressed in Figure 4. A more formal test is achieved by the *kitchen sink regression*,

$$\log(r_{1,t}^2/RV_{2,t}) = \alpha + \beta' \mathbf{Z}_t + u_t, \quad (13)$$

where $\mathbf{Z}_t$ contains *instruments* that must be uncorrelated with ε_t and ν_t .⁷ The validity of Assumption 1 is now questioned by testing the hypothesis, $H_0 : \beta = \mathbf{0}$. While the instruments must be uncorrelated with ν_t and ε_t , it is highly desirable that the instruments be correlated with IV_t , because this correlation is the source of the power of the test.

Lee, Ready & Seguin (1994) analyzed the futures market and found a sharp drop in volatility subsequent to the close of the NYSE, but an increase in volatility at the closing time of the futures market. Chang, Jain & Locke (1995) analyzed trading halts at the NYSE, and found such to increase volatility over the subsequent trading day, and Gallo (2001) studied the impact of the overnight return on volatility the subsequent trading day.

⁶The few days where $r_{1,t} = 0$ were omitted from the scatter plots (to avoid taking the logarithm to zero).

⁷We thank Tim Bollerslev for suggesting the kitchen sink regression. This name for this regression reflects that a large number of variable can be “tossed in” as regressors.

Lagged variables are expected to be uncorrelated with the “measurement” errors, ν_t and ε_t , because these errors are specific to day t , see Barndorff-Nielsen & Shephard (2002a) for an excellent discussion of the properties of $RV_t^{(m)} - IV_t$ in the absence of market microstructure noise. So lagged variables define a large class of potential instruments that could be used in the regression (13). Daily volatility is known to be highly persistent, so

$$\log(RV_{2,t-1}) = \log(IV_{t-1}) + \log[c_1(1 + \varepsilon_{t-1})],$$

is expected to be a strong instrument, due to the high correlation between $\log(IV_{t-1})$ and $\log(IV_t)$.

Deterministic variables define another class of potential instruments that satisfy the necessary moment conditions concerning ν_t and ε_t . Thus we define the dummy variables, $D_{\text{mon},t}$, $D_{\text{tue},t}$, $D_{\text{wed},t}$, $D_{\text{thu},t}$, and $D_{\text{fri},t}$, that are indicator functions for the five weekdays, i.e.,

$$D_{\text{mon},t} \equiv \begin{cases} 1 & \text{if day } t \text{ is a Monday,} \\ 0 & \text{otherwise,} \end{cases}$$

with similar definitions for $D_{\text{tue},t}$, $D_{\text{wed},t}$, $D_{\text{thu},t}$, and $D_{\text{fri},t}$. The variation in average volatility across weekdays motivates these variables as instruments. In the regression, (13), we can only include four of these five dummy variable to avoid colinearity problems (since $D_{\text{mon},t} + D_{\text{tue},t} + D_{\text{wed},t} + D_{\text{thu},t} + D_{\text{fri},t} = 1$). So our empirical results are based on

$$\mathbf{Z}_t \equiv (\log(RV_{2,t-1}), D_{\text{mon},t}, D_{\text{tue},t}, D_{\text{wed},t}, D_{\text{thu},t})'$$

and we estimate α and β by least squares. The p -values from the kitchen sink regressions are presented in Panel A of Table 3. All reported p -values are heteroskedastic robust, see White (1980). These regressions are based on a truncated sample that excluded the days where $r_{1,t} = 0$. Two of the 30 tests suggest a rejection of $H_0 : \beta = \mathbf{0}$ at the 5% significance level. Naturally, some rejections are to be expected when 30 hypothesis are being tested. Since only one of the test are more than marginally significant, we do not find the evidence against Assumption 1 to be much stronger than is expected by pure chance.

[Table 3 about here]

A second assumption that is critical for the construction of RV_t^* is that the estimator is conditionally unbiased, $E_\sigma[RV_t(\omega)] = IV_t$. This requirement is clearly much stronger than the unconditional version $E[RV_t(\omega)] = E[IV_t]$ that is imposed in our estimation procedure. So we investigate the validity of $E_\sigma[RV_t(\omega)] = IV_t$ by partitioning the sample in two ways to check if there are any obvious anomalies with respect to these partitions of our sample. Ideally we would partition the sample with respect to the quantiles of the latent variable $IV_{2,t}$. Since these are unobserved we take a slightly different approach and use the quantiles of $RV_{2,t-1}$ as instruments. These are valid instruments under the fairly realistic assumption that the measurement

error $RV_{2,t-1} - IV_{2,t-1}$ is uncorrelated with the measurement error $RV_{2,t} - IV_{2,t}$. Further, these quantiles are expected to be highly correlated with those of $IV_{2,t}$, because of the high persistence of volatility that is observed in practice. In other words, this approach is likely to generate strong and valid instruments that leads to a powerful specification check of the conditional condition. Specifically we define $Q_{j,t} \equiv 1_{\{RV_{2,t-1} \in I_j\}}$, $j = 1, \dots, 5$ and $t = 1, \dots, n$, where I_1, I_2, I_3, I_4 , and I_5 is the segmentation of $RV_{2,t-1}$'s support, using the 20%, 40%, 60%, and 80% empirical quantiles.

Volatility may also be subject to some systematic variation across weekdays. Issues related to the weekend (close-to-open) return is of particular interest as we discussed after Assumption 1. So we check whether there is any systematic measurement error in RV_t^* , in relation to weekdays.

Thus, by calculating empirical ratios

$$\frac{\sum_{t=1}^n RV_t^* Q_{j,t}}{\sum_{t=1}^n (r_{1,t}^2 + RV_{2,t}^{(13)}) Q_{j,t}}, \quad \text{and} \quad \frac{\sum_{t=1}^n RV_t^* D_{j,t}}{\sum_{t=1}^n (r_{1,t}^2 + RV_{2,t}^{(13)}) D_{j,t}}, \quad (14)$$

and verifying that these are close to one, we are implicitly checking the identities,

$$\begin{aligned} E(RV_t^* | Q_{j,t}) &= E(IV_t | Q_{j,t}), & \text{for } j = 1, \dots, 5, & \quad \text{and} \\ E(RV_t^* | D_{j,t}) &= E(IV_t | D_{j,t}), & \text{for } j = \text{mon}, \dots, \text{fri}. \end{aligned}$$

The ratios in (14) were found to be sensitive to outliers. So the empirical ratios we report in Panel B of Table 3 are for the truncated sample, see footnote 4, that consists of about 99% of our observations. Although we do not report standard errors for these ratios, it is comforting to see that all ratios are fairly close to unity. So there does not appear to be any systematic anomalies with respect to these instruments.

Finally we conduct a formal test of the conditional unbiasedness that is based on a second kitchen sink regression,

$$\log[RV_t^* / (r_{1,t}^2 + RV_{2,t}^{(13)})] = \alpha + \beta' \mathbf{Z}_t + u_t, \quad (15)$$

where we use the full sample and employ the same vector of instruments as in the first kitchen sink regression. We estimate α and β by least squares using the full sample for each of the 30 equities, and the heteroskedasticity robust p -values for tests of $H_0 : \beta = \mathbf{0}$ are given in the last column of Table 3. Three of these tests are found to be significant at the 5% significance level. The very small p -values for HD and XOM raise some concerns about the properties of RV_t^* for these two assets, although these may be caused by a few outliers. Our overall impression is that RV_t^* is generally a reasonable estimator of IV_t .

5 Summary

In this paper, we have analyzed the problem of estimating the daily integrated variance, IV_t , in the common situation where high-frequency data are unavailable for part of the day – the inac-

tive close-to-open period. We referred to this situation as one with intermittent high-frequency data. Within this framework, we studied some existing estimators of daily volatility, IV_t , and proposed a new estimator that combines the squared over-night return, $r_{1,t}^2$, and a realized variance type quantity for the active period, $RV_{2,t}$. We established assumptions that justify these estimators and showed that the optimal linear combination of $r_{1,t}^2$ and $RV_{2,t}$ is empirically solvable, once we restrict our attention to conditionally unbiased estimators. We discussed additional assumptions that makes it possible to incorporate more information in the estimation of daily volatilities. A smoothing across equities or across days can improve the precision of our daily estimators under suitable assumptions.

We applied the results to high-frequency data for the 30 stocks of the Dow Jones Industrial Average, and have constructed four years of daily volatility estimates for these equities. In our empirical analysis we tested the assumptions that were critical for our estimators, and the results were generally favorable to the estimators. While the estimated optimal linear combination of $r_{1,t}^2$ and $RV_{2,t}$ generally had a disproportionately larger weight on the latter, the squared overnight return, $r_{1,t}^2$, was always given some (often substantial) weight. Thus, despite the fact that $r_{1,t}^2$ is a great deal more noisy than is $RV_{2,t}$, the squared overnight return, $r_{1,t}^2$, does contain valuable information about daily volatility, over and above the information contained in $RV_{2,t}$.

A Appendix of Proofs

Proof of Theorem 1. Combining *i* and *ii* of Assumption 1 yields $E_\sigma(\delta RV_{2,t}) = E_\sigma(IV_{2,t}/\delta_o) = IV_t$, which proves the first result of the theorem. The unconditional version of this identity yields $E(RV_{2,t}) = E(IV_t)/\delta$. It is assumed that $dp^*(t) = \mu dt + \sigma(t)dw(t)$, such that $r_t = \mu + \int_{t-1}^t \sigma(s)dw(s)$ and $r_t|\sigma(t) \sim N(\mu, IV_t)$. So by Assumption 2 we have that $n^{-1} \sum_{t=1}^n RV_{2,t} \xrightarrow{P} E(RV_{2,t})$, $\bar{r} \xrightarrow{P} E(r_t) (= \mu)$ and $n^{-1} \sum_{t=1}^n (r_t - \bar{r})^2 \xrightarrow{P} \text{var}(r_t) = E(IV_t)$. So it follows that $\hat{\delta} = n^{-1} \sum_{t=1}^n (r_t - \bar{r})^2 / n^{-1} \sum_{t=1}^n RV_{2,t} \xrightarrow{P} \delta$, such that $(\hat{\delta} - \delta) RV_{2,t} = o_p(1)O_p(1) = o_p(1)$. ■

Proof of Lemma 4. We need to show that $E_\sigma(\omega_1 r_{1,t}^2 + \omega_2 RV_{2,t}) = IV_t$ for all (ω_1, ω_2) that satisfy $\mu_0 = \omega_1 \mu_1 + \omega_2 \mu_2$. First, we note that $E_\sigma(IV_{1,t}) = E_\sigma(IV_t - IV_{2,t}) = (1 - \delta_o)IV_t$ by Assumption 1.i. Next, by Assumption 1.ii-iii we have that

$$E_\sigma(r_{1,t}^2) = E_\sigma(IV_{1,t})/\delta_{b_1} = \delta_1 IV_t, \quad \text{and} \quad (16)$$

$$E_\sigma(RV_{2,t}) = E_\sigma(IV_{2,t})/\delta_{b_2} = \delta_2 IV_t. \quad (17)$$

where $\delta_1 \equiv (1 - \delta_o)/\delta_{b_1}$ and $\delta_2 \equiv \delta_o/\delta_{b_2}$, and combining (16) and (17) leads to the identity

$$E_\sigma[RV_t(\omega)] = \omega_1 \delta_1 IV_t + \omega_2 \delta_2 IV_t = (\omega_1 \delta_1 + \omega_2 \delta_2) IV_t. \quad (18)$$

Finally, by the assumption, $\mu_0 = \omega_1 \mu_1 + \omega_2 \mu_2$, and the unconditional expectation of (18), we have that

$$E[RV_t(\omega)] = \begin{cases} \omega_1 \mu_1 + \omega_2 \mu_2 = \mu_0 \\ (\omega_1 \delta_1 + \omega_2 \delta_2) \mu_0, \end{cases}$$

which proves that $\omega_1\delta_1 + \omega_2\delta_2 = 1$. Thus, (18) simplifies to $E_\sigma[RV_t(\omega)] = IV_t$, which completes the proof. ■

Proof of Theorem 5. Let $\hat{\theta}$ and $\tilde{\theta}$ be two unbiased estimators for θ . Then $\lambda^*\hat{\theta} + (1 - \lambda^*)\tilde{\theta}$ is the linear combination with the smallest variance where $\lambda^* = [\text{var}(\tilde{\theta}) - \text{cov}(\hat{\theta}, \tilde{\theta})]/[\text{var}(\hat{\theta}) + \text{var}(\tilde{\theta}) - 2\text{cov}(\hat{\theta}, \tilde{\theta})]$. The result now follows by setting $\hat{\theta} = \frac{\mu_0}{\mu_1}RV_{2,t}$ and $\tilde{\theta} = \frac{\mu_0}{\mu_2}r_{1,t}^2$ and using simple variance and covariance formulae. ■

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Table 1: Equities included in our empirical analysis.

Symbol	Name	Exchange	Mean #trans.	Mean #quotes	#Days
AA	ALCOA INC	NYSE	2180	4419	986
AXP	AMERICAN EXPRESS	NYSE	2837	5421	985
BA	BOEING COMPANY	NYSE	2516	5160	985
C	CITIGROUP	NYSE	4096	6604	984
CAT	CATERPILLAR INC	NYSE	2144	4760	984
DD	DU PONT DE NEMOURS	NYSE	2502	5284	984
DIS	WALT DISNEY	NYSE	2987	6005	984
EK	EASTMAN KODAK	NYSE	1621	3737	984
GE	GENERAL ELECTRIC	NYSE	4376	6936	984
GM	GENERAL MOTORS	NYSE	2401	4884	984
HD	HOME DEPOT INC	NYSE	3422	5620	984
HON	HONEYWELL	NYSE	2127	4369	984
HPQ	HEWLETT-PACKARD	NYSE	2960	5432	981
IBM	INT. BUSINESS MACHINES	NYSE	3943	7169	984
INTC	INTEL CORP	NASDAQ	12937	17430	984
IP	INTERNATIONAL PAPER	NYSE	2088	4492	984
JNJ	JOHNSON AND JOHNSON	NYSE	2941	5188	984
JPM	J.P. MORGAN	NYSE	3331	5934	984
KO	COCA-COLA	NYSE	2561	4786	984
MCD	MCDONALDS	NYSE	2363	4647	984
MMM	MINNESOTA MNG MFG	NYSE	2531	5287	984
MO	PHILIP MORRIS	NYSE	2981	5498	984
MRK	MERCK	NYSE	2967	5084	984
MSFT	MICROSOFT	NASDAQ	12984	17801	984
PG	PROCTER & GAMBLE	NYSE	2886	5932	984
SBC	SBC COMMUNICATIONS	NYSE	2935	5404	984
T	AT & T CORP	NYSE	2206	4412	984
UTX	UNITED TECHNOLOGIES	NYSE	2321	5007	984
WMT	WAL-MART STORES	NYSE	3130	5273	984
XOM	EXXON MOBIL	NYSE	3612	6460	983

The table lists symbols and names of the 30 equities of the DJIA that are used in our empirical analysis. For each of the equities, we list the exchange that data were extracted from, the average number of transactions and quotes per day, and the number of trading days in our sample.

Table 2: Optimal measure of daily volatility: Empirical estimates of components.

Asset	$\hat{\mu}_0$	$\hat{\mu}_1$	$\hat{\mu}_2$	$\hat{\mu}_2/\hat{\mu}_1$	$\hat{\eta}_1^2$	$\hat{\eta}_2^2$	$\hat{\eta}_2^2/\hat{\eta}_1^2$	$\frac{\hat{\eta}_{12}}{\hat{\eta}_1\hat{\eta}_2}$	$\hat{\varphi}$	$\hat{\omega}_1^*$	$\hat{\omega}_2^*$
AA	4.694	1.080	3.819	3.535	5.410	9.587	1.772	0.325	0.978	0.094	1.203
AXP	4.249	0.924	3.394	3.674	4.706	18.17	3.861	0.292	0.867	0.614	1.085
BA	4.117	0.737	3.523	4.782	2.206	11.31	5.130	0.236	0.887	0.629	1.037
C	3.924	1.024	3.160	3.087	5.257	12.25	2.330	0.358	0.924	0.291	1.148
CAT	3.312	0.582	2.678	4.603	1.506	4.480	2.976	0.262	0.955	0.255	1.181
DD	2.960	0.608	2.532	4.166	1.433	5.320	3.712	0.287	0.914	0.418	1.069
DIS	5.011	1.226	4.078	3.326	6.273	14.04	2.238	0.323	0.937	0.256	1.152
EK	4.693	0.888	3.733	4.205	7.924	14.74	1.861	0.284	0.986	0.075	1.239
GE	3.810	0.974	2.952	3.030	4.822	9.912	2.055	0.287	0.908	0.362	1.171
GM	3.534	0.696	2.724	3.912	2.123	6.949	3.274	0.194	0.880	0.610	1.141
HD	4.760	1.180	3.596	3.046	8.830	13.40	1.517	0.255	0.937	0.254	1.240
HON	5.626	1.307	4.488	3.433	9.521	22.87	2.402	0.317	0.934	0.285	1.170
HPQ	7.926	2.501	5.754	2.301	53.24	34.27	0.644	0.253	0.965	0.112	1.329
IBM	3.097	0.825	2.242	2.717	3.938	5.889	1.495	0.260	0.912	0.331	1.260
INTC	8.393	2.482	6.112	2.463	36.50	32.26	0.884	0.146	0.913	0.294	1.254
IP	2.995	0.507	2.642	5.216	0.876	4.912	5.608	0.220	0.894	0.625	1.014
JNJ	2.271	0.494	1.854	3.753	1.031	4.490	4.356	0.311	0.858	0.650	1.051
JPM	5.485	1.420	4.215	2.968	11.82	22.61	1.913	0.323	0.927	0.281	1.207
KO	2.221	0.468	1.828	3.906	1.048	2.741	2.616	0.313	0.954	0.218	1.159
MCD	3.546	0.788	2.953	3.746	2.767	7.796	2.818	0.404	0.976	0.107	1.172
MMM	2.156	0.314	1.908	6.075	0.465	3.952	8.494	0.263	0.894	0.730	1.010
MO	2.763	0.596	2.161	3.624	3.133	7.482	2.388	0.250	0.922	0.360	1.179
MRK	2.909	0.667	2.363	3.544	2.621	7.199	2.747	0.266	0.903	0.425	1.111
MSFT	4.047	0.977	3.299	3.376	5.707	10.47	1.835	0.205	0.921	0.327	1.130
PG	1.634	0.281	1.456	5.188	0.376	2.010	5.352	0.288	0.925	0.436	1.038
SBC	4.633	0.921	3.897	4.231	4.375	16.03	3.663	0.331	0.939	0.304	1.117
T	5.424	1.343	4.162	3.099	8.060	15.84	1.966	0.313	0.931	0.277	1.214
UTX	3.092	0.555	2.596	4.678	1.328	6.971	5.248	0.288	0.897	0.575	1.068
WMT	2.863	0.558	2.382	4.271	1.251	5.695	4.552	0.281	0.887	0.579	1.066
XOM	2.098	0.483	1.799	3.726	0.840	3.634	4.326	0.398	0.897	0.449	1.045

The table contains the empirical estimates that are used to compute the optimal measure of daily volatility. The importance factor, φ , is close to one in all cases which causes the realized variance for the active period, $RV_{2,t}$ to be scaled upwards ($\omega_1 > 1$), whereas the squared overnight return, $r_{1,t}$, is scaled downward ($\omega_2 < 1$). The estimates of φ , ω_1^* , and ω_2^* are defined from (8) and (9) and the other estimates are defined by:

$$\begin{aligned}\hat{\mu}_0 &\equiv \frac{1}{n} \sum_{t=1}^n (r_{1,t}^2 + RV_{2,t}^{(13)}), & \hat{\mu}_1 &\equiv \frac{1}{n} \sum_{t=1}^n r_{1,t}^2, & \hat{\mu}_2 &\equiv \frac{1}{n} \sum_{t=1}^n RV_{2,t}, \\ \hat{\eta}_1^2 &\equiv \frac{1}{n} \sum_{t=1}^n (r_{1,t}^2 - \hat{\mu}_1)^2, & \hat{\eta}_2^2 &\equiv \frac{1}{n} \sum_{t=1}^n (RV_{2,t} - \hat{\mu}_2)^2, & \hat{\eta}_{12} &\equiv \frac{1}{n} \sum_{t=1}^n RV_{2,t} (r_{1,t}^2 - \hat{\mu}_1).\end{aligned}$$

Table 3: Kitchen sink regressions.

Symbol	Panel A: Assumption 1	Panel B: Conditional Unbiasness					Mon	Tue	Wed	Thu	Fri	<i>p</i> -value
	<i>p</i> -value	q1	q2	q3	q4	q5						
AA	0.152	0.98	0.97	0.92	1.03	1.03	0.97	0.97	0.98	1.02	1.06	0.187
AXP	0.850	0.91	0.95	1.01	0.98	1.04	0.96	1.00	1.05	1.03	0.96	0.630
BA	0.832	0.94	0.96	0.99	1.04	1.02	0.97	1.00	1.02	0.98	1.03	0.485
C	0.060	1.02	0.98	0.99	1.02	1.00	0.93	1.02	0.98	1.09	0.99	0.097
CAT	0.490	1.03	0.96	1.06	1.03	0.99	0.92	1.01	0.98	1.04	1.04	0.485
DD	0.869	0.96	0.95	1.00	1.04	1.02	0.94	0.98	0.99	1.01	1.07	0.417
DIS	0.828	1.06	1.01	0.90	0.98	1.03	0.96	1.00	1.00	1.03	1.01	0.734
EK	0.247	0.96	1.00	1.04	1.02	0.99	1.00	1.02	0.98	0.99	1.02	0.080
GE	0.053	1.00	1.04	1.01	0.96	1.01	0.96	1.01	1.01	1.03	0.98	0.147
GM	0.898	1.02	0.98	0.92	0.97	1.05	0.93	0.99	1.02	1.05	1.01	0.111
HD	0.001	1.03	0.96	0.91	1.06	1.00	0.91	0.94	1.07	1.03	1.07	0.005
HON	0.215	0.99	1.02	0.95	1.03	0.99	0.95	0.97	1.05	1.01	1.02	0.570
HPQ	0.598	0.91	1.01	0.97	1.05	0.99	0.98	1.04	0.93	0.97	1.13	0.434
IBM	0.125	1.06	0.93	1.04	1.00	1.01	1.00	1.02	1.00	0.98	0.99	0.666
INTC	0.468	0.80	0.93	0.95	1.04	1.06	0.95	1.04	0.97	1.05	1.00	0.521
IP	0.883	0.94	0.95	0.99	1.00	1.05	0.95	0.98	1.03	1.03	1.02	0.175
JNJ	0.539	0.91	0.96	0.97	1.02	1.04	0.95	0.95	0.99	1.05	1.07	0.054
JPM	0.412	0.99	1.00	0.93	0.98	1.03	0.91	1.03	1.00	1.03	1.05	0.052
KO	0.733	1.01	1.01	1.01	0.96	1.01	1.03	1.01	0.96	1.01	0.99	0.955
MCD	0.042	0.90	0.97	1.08	1.01	1.03	0.93	1.01	1.06	1.02	0.98	0.767
MMM	0.068	0.95	0.98	1.05	1.00	1.02	0.95	0.98	1.03	1.01	1.02	0.316
MO	0.113	1.07	1.02	1.04	0.96	1.00	1.02	0.95	0.98	1.01	1.05	0.054
MRK	0.722	0.91	0.97	0.96	1.01	1.04	0.91	0.96	1.06	1.09	0.98	0.053
MSFT	0.109	1.00	0.94	1.02	0.96	1.04	0.96	0.99	0.97	1.09	0.99	0.584
PG	0.984	0.94	0.98	0.98	1.00	1.02	0.95	0.99	1.00	1.03	1.02	0.777
SBC	0.475	0.98	0.98	0.91	0.93	1.07	0.97	0.98	1.02	0.99	1.05	0.228
T	0.155	0.94	1.00	0.92	1.01	1.03	0.96	0.93	1.07	1.00	1.05	0.035
UTX	0.175	1.00	0.97	1.01	1.00	1.01	0.94	1.02	1.04	0.97	1.03	0.096
WMT	0.191	0.94	0.97	0.98	1.01	1.02	0.97	0.97	1.01	1.01	1.04	0.102
XOM	0.871	0.92	0.96	1.00	1.00	1.03	1.01	0.98	0.96	1.00	1.05	<.001

Panel A contains the p -values from the first kitchen sink regression (13) and these yield little evidence against Assumption 1. The first ten columns of Panel B contain the bias ratios that are defined in (14). The first five columns are based on empirical proxies for the quintiles of IV_t , whereas the sample are segmented according to weekday in the next five columns. The last column contains the p -values from the second kitchen sink regression (15). We report heteroskedasticity robust p -values, and those that are significant at the 5% level are identified by bold font.

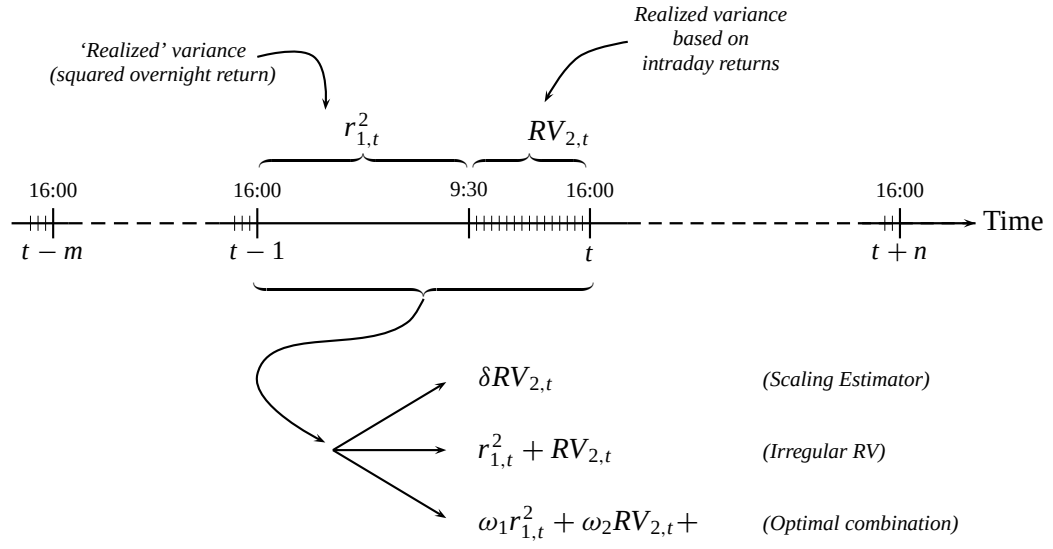


Figure 1: Illustration of setup

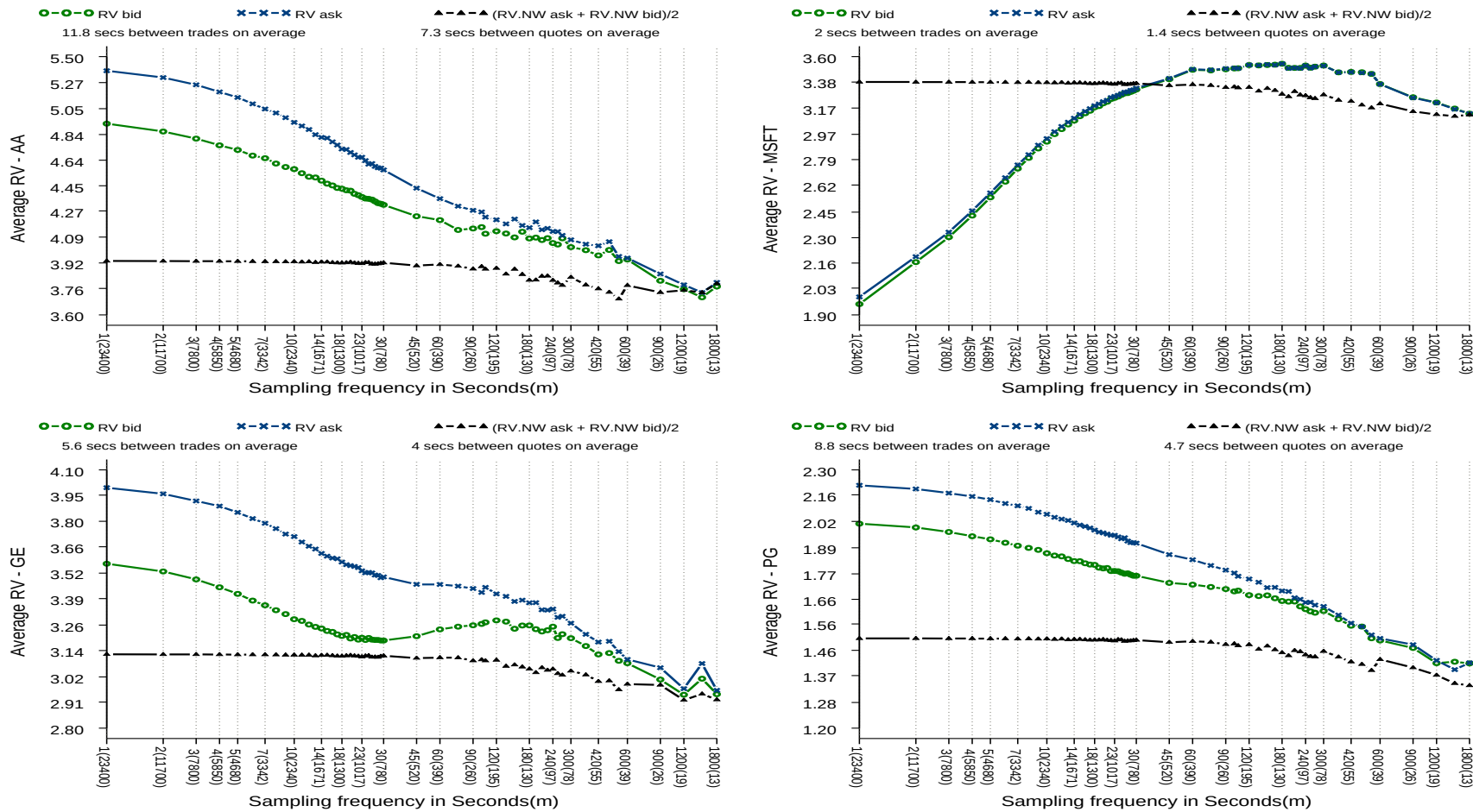


Figure 2: This figure contains three volatility signature plots. Circles refer to the $RV^{(m)}$ based on bid quotes; crosses refer to the $RV^{(m)}$ based on ask quotes, and triangles identify the Newey-West modified realized variance, $RV_{NW,t}^{(m)}$. The last measure is the average of two $RV_{NW,t}^{(m)}$ s – one based on bid quotes and one based on ask quotes.

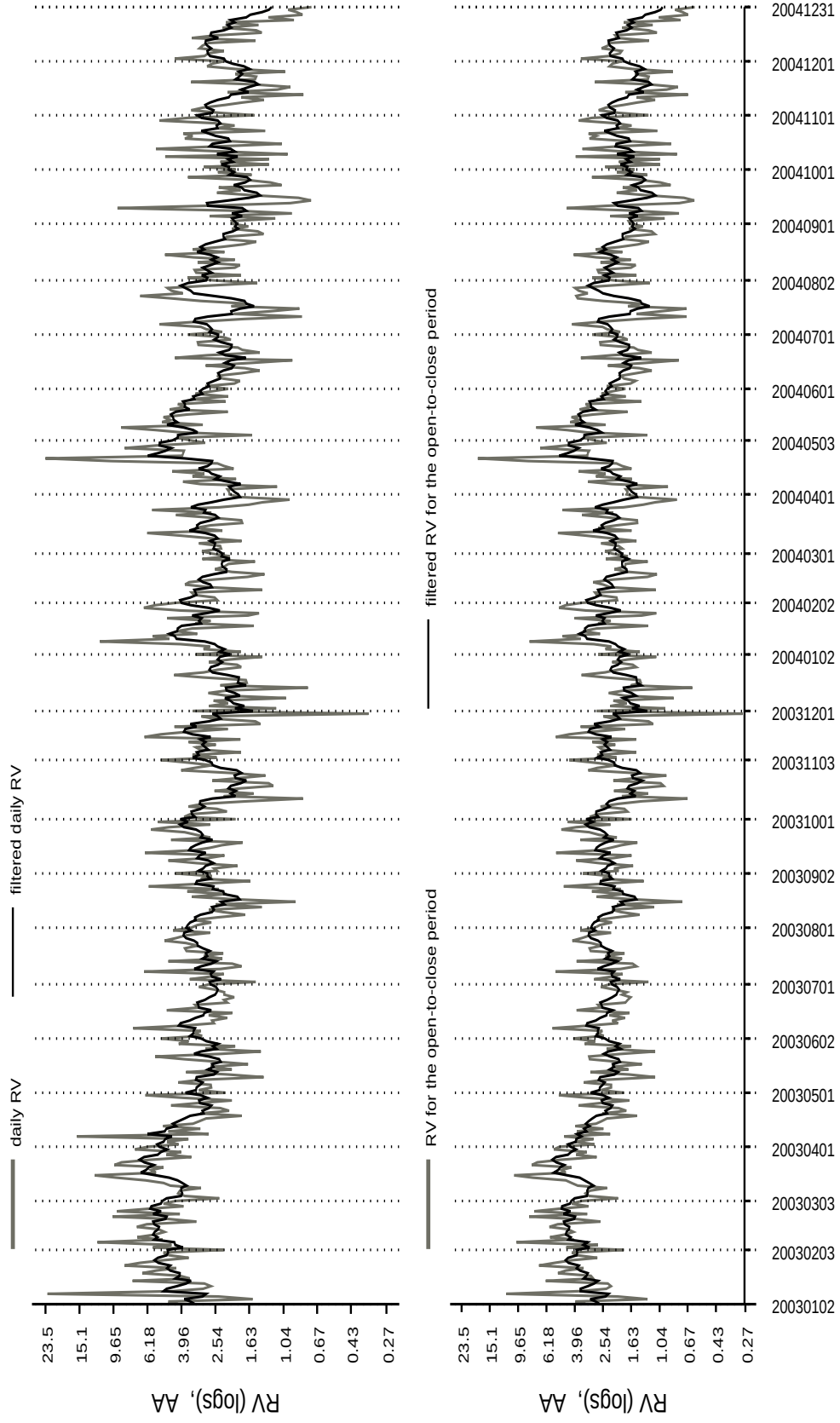


Figure 3: Time series plots of RV_t^* , $\widetilde{RV}_t^*$, $RV_{2,t}$, and $\widetilde{RV}_{2,t}$.

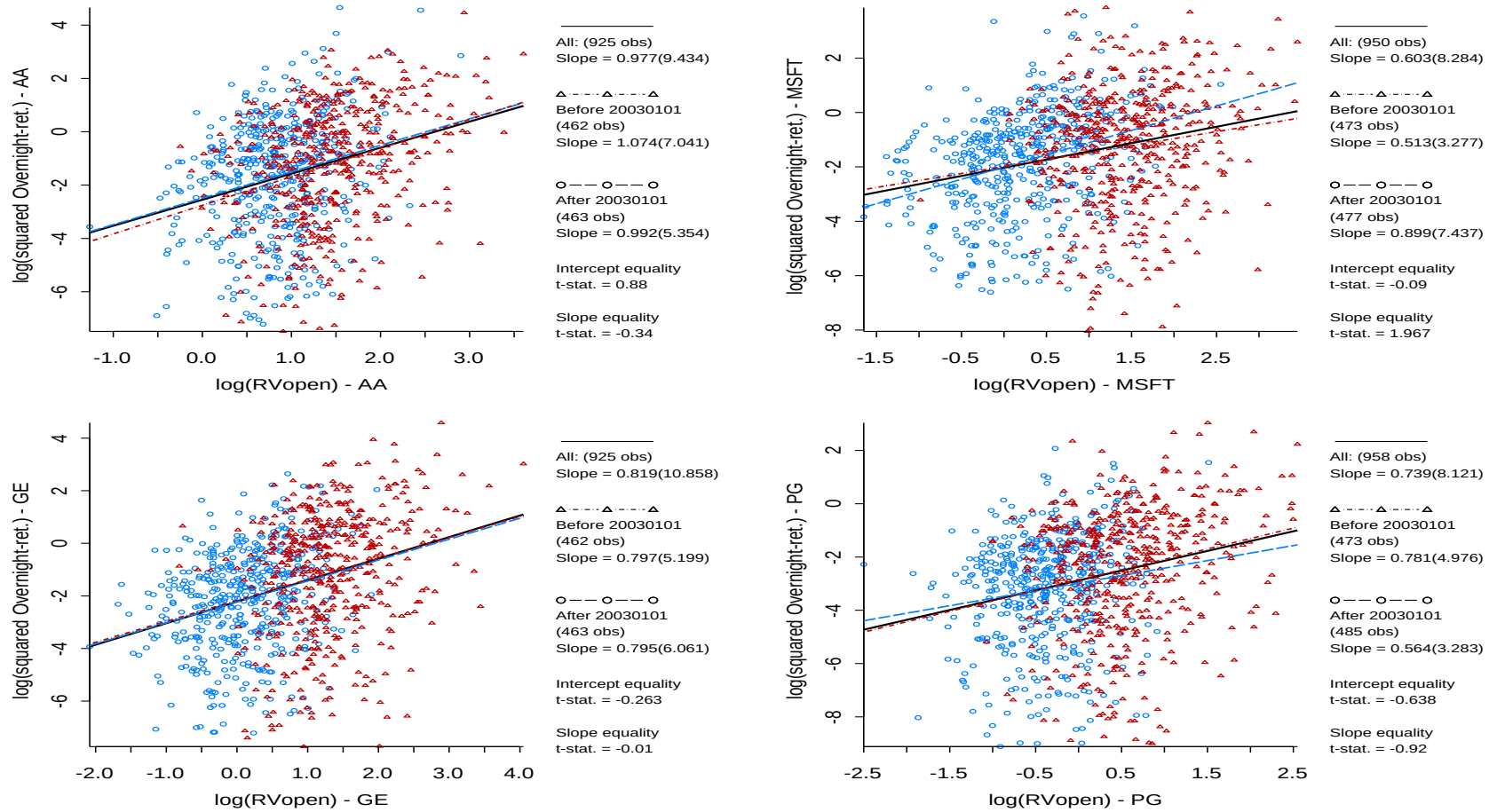


Figure 4: The figure contains scatter plots of $\log(r_{1,t}^2)$ against $\log(RV_{2,t})$. Triangles identify observations for 2001 and 2002, whereas circles identify observations for 2003 and 2004. The three lines in each of the subplots are fitted regression lines for the two subsamples and the full sample. The scatter plots provide a visual inspection of Assumption 1, because the regression lines ought to coincide under this assumption. The reported t -statistics correspond to t -tests of identical slope and intercept, and all are found to be insignificant.